

Total Questions: 35

LEVEL – II

Set - B

Time: 1.5 hr.

Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. The displacement–time graph of a moving particle is a curve whose slope is positive but continuously decreasing with time. What can be definitely concluded about the particle's motion?

- (A) It is moving in the negative direction.
- (B) It is moving with negative acceleration (retardation).
- (C) It is moving with increasing acceleration.
- (D) It is moving with constant velocity.

2. In high-altitude mountaineering, climbers often find it difficult to cook food in open pots because water boils at a lower temperature (e.g., 90°C). Which statement best explains this phenomenon?

- (A) The oxygen levels are too low to sustain a hot flame.
- (B) Decreased atmospheric pressure allows liquid molecules to escape into the vapor phase with less thermal energy.

- (C) The cold temperature of the mountains absorbs the heat from the water too quickly.
- (D) Gravity is weaker at high altitudes, making it easier for steam to rise.
3. **A mutation occurs in a cell that prevents the Rough Endoplasmic Reticulum (RER) from attaching ribosomes to its surface. How will this specifically impact the cell's proteome (the entire set of proteins expressed)?**
- (A) The cell will stop producing all proteins, leading to immediate cell death.
- (B) The cell will continue to produce cytoplasmic proteins via free ribosomes, but will fail to produce membrane-bound or secretory proteins.
- (C) The cell will convert all its RER into SER to compensate for the loss of protein synthesis.
- (D) Ribosomes will move to the mitochondria to continue the synthesis of secretory proteins.
4. **A constant force acts on a body. In two different inertial frames, displacement is different. The work done.**
- (A) Remains same
- (B) Becomes zero
- (C) Becomes different
- (D) Depends only on force
5. **Brass is an alloy consisting of approximately 60% Copper and 40% Zinc. Even though it is a solid-solid mixture and the components can be varied in different proportions, it is classified as a homogeneous mixture because:**

- (A) Copper and Zinc react chemically to form a new compound.
- (B) The constituents cannot be separated by physical methods like filtration.
- (C) It exhibits a uniform composition and properties throughout the sample.
- (D) Zinc is always soluble in Copper regardless of temperature.
6. **Why do plant cells not burst when placed in a highly hypotonic solution like pure rainwater, whereas animal cells (like RBCs) undergo lysis (bursting) under the same conditions?**
- (A) Plant cells have a higher internal osmotic pressure that repels incoming water molecules.
- (B) Plant cells possess contractile vacuoles that pump out water as fast as it enters the cell.
- (C) The presence of a rigid cellulose cell wall in plants exerts turgor pressure against the swelling cytoplasm.
- (D) The plasma membrane of a plant cell is thicker and more elastic than that of an animal cell.
7. **Which of the following characteristics in a displacement–time (s–t) graph is physically impossible for a real object moving in one dimension?**
- (A) A curve whose slope continuously decreases and becomes negative.
- (B) A vertical line segment parallel to the displacement axis.
- (C) A horizontal line segment parallel to the time axis.
- (D) A sharp, angled corner where the slope abruptly changes

8. Alum, often used in water purification, contains Aluminium (Al , Valency 3) and Sulphate ions (SO_4^{-2}), Valency 2). When calculating the molecular weight of Aluminium Sulphate, $\text{Al}_x (\text{SO}_4)_y$, how does the valency affect the total mass contribution of Sulphur?

- (A) Valency of Al (3) becomes the subscript for SO_4 , meaning 3 atoms of Sulphur are counted.
- (B) Valency of SO_4 (2) becomes the subscript for SO_4 , meaning 2 atoms of Sulphur are counted.
- (C) Valency has no impact on the number of atoms, only on the charge.
- (D) The subscripts are $x=1$, $y=1$, so only 1 atom of Sulphur is counted.

9. Which of the following correctly matches the label with its unique characteristic?



- (A) **P**: Selectively permeable and made of phospholipids.
- (B) **Q**: Maintains turgidity and stores cell sap.
- (C) **R**: Site of aerobic respiration and ATP production.
- (D) **S**: Contains chlorophyll and performs carbon fixation.

10. A body is projected upward with speed equal to half the escape velocity. The maximum height reached from Earth's surface is:

- (A) R
- (B) $R/2$
- (C) $R/3$
- (D) $R/4+$

11. Follow the information given below:

Alpha particle experiment



Most pass → empty space

Few deflect →?

Very few bounce → dense centre

What does “few deflect” indicate?

- (A) Presence of electrons
- (B) Presence of nucleus
- (C) Atom is positive
- (D) Atom is negative

12. A botanist is studying a high-altitude plant that experiences frequent heavy winds. Upon microscopic examination of the leaf stalks (petioles), she finds a tissue characterized by cells with irregularly thickened corners made of cellulose and pectin, with very little intercellular space. Which of the following best describes this tissue and its benefit to the plant?

- (A) Sclerenchyma; it provides rigid, stone-like hardness to prevent the leaf from snapping.
- (B) Parenchyma; it stores large amounts of water to keep the leaf hydrated during dry winds.
- (C) Collenchyma; it provides mechanical support and flexibility, allowing the petiole to bend without breaking.
- (D) Aerenchyma; it creates buoyancy so the leaves can float in the thin mountain air.

13. A group of researchers discovered a mutant variety of a desert plant where the 'suberin' deposition in the bark and the 'cutin' layer on the epidermis are significantly reduced. What is the most likely fate of this mutant plant in its natural habitat?

- (A) The plant will grow much faster because it can absorb more CO₂ from the atmosphere.
- (B) The plant's internal tissues will stay cooler because heat can escape more easily.
- (C) The plant will become more resistant to fungal infections as the skin is thinner.
- (D) The plant will wilt and die quickly due to excessive, uncontrolled loss of water.

14. In a fish culture system, a farmer uses a "Composite Fish Culture" method with five species: Catla, Rohu, Mrigal, Silver Carp, and Grass Carp. After a few months, he observes that the Grass Carp are not growing well, even though the water quality is excellent. What is the most likely reason for the poor growth of the Grass Carp?

- (A) They are being outcompeted for food by Catla, which also feeds on the surface.
- (B) There is a lack of aquatic weeds and vegetation in the pond for them to consume.
- (C) They are highly sensitive to the presence of Silver Carp, which produces growth-inhibiting hormones.
- (D) The pond is too deep, preventing the Grass Carp from reaching the bottom to feed.

15. An "Organic Farmer" refuses to use synthetic pesticides to control a pest infestation in his Mustard crop. Instead, he introduces ladybugs into the field and uses a neem-based spray. Which principles of sustainable agriculture is he applying?

- (A) Genetic engineering and chemical control.
- (B) Monoculture and intensive irrigation.
- (C) Biological control and use of bio-pesticides.
- (D) Eutrophication and soil salinization.

Section - 2

16. Earth's equatorial radius is 21 km larger than polar radius. If $g = \frac{GM}{R^2}$, then

- (A) $g_p > g_e$
- (B) $\frac{g_e}{g_p} = \frac{R_p^2}{R_e^2}$
- (C) A person weighs more at poles than equator
- (D) $g_e > g_p$

17. Which of the following electronic configurations represents a violation of Hund's Rule but NOT a violation of the Pauli Exclusion Principle?

- (A) An orbital containing two electrons with the same spin (parallel spins).
- (B) A carbon atom with both electrons paired in the same orbital with opposite spins.
- (C) A nitrogen atom with two electrons in one orbital and one in another.
- (D) An electron entering a orbital before the subshell is full.

18. Mitochondria and Chloroplasts are often referred to as "semi-autonomous organelles." Which of the following biological facts support this classification?

- (A) They possess their own circular DNA independent of the nucleus.
- (B) They can synthesize some of their own proteins using their own ribosomes.
- (C) They are capable of duplicating themselves through a process similar to fission.
- (D) They can survive and function indefinitely if removed from the host cell.

19. A force varies with displacement as $F = 4x - 2$ (in N), where x is in meters. A particle moves from $x = 0$ to $x = 2$.

- (A) Total work done is 4 J
- (B) Work done is negative in some part of motion
- (C) Net work done is equal to area under $F-x$ graph
- (D) Work done becomes zero if motion is reversed

20. If you have 0.5 'gram molecules' of Water (H_2O), how many Hydrogen atoms are present?

- (A) 6.022×10^{23} atoms
- (B) 1.0 atom
- (C) 1.204×10^{24} atoms
- (D) 3.011×10^{23} atoms

21. Which of the following are functions performed by the Smooth Endoplasmic Reticulum (SER) in various types of animal cells?

- (A) Synthesis of lipids and steroid hormones like estrogen and testosterone.

(B) Detoxification of poisons and drugs, particularly in liver cells.

(C) Sorting and tagging of proteins with carbohydrates for export.

(D) Storage of calcium ions required for muscle contraction.

22. Complex permanent tissues in plants are "complex" because they are made of more than one type of cell working together. Identify the CORRECT facts about Xylem and Phloem:

(A) Xylem consists of tracheids, vessels, xylem parenchyma, and xylem fibers.

(B) In Phloem, the only living component is the Phloem fiber.

(C) Vessels and tracheids are dead cells that allow for the vertical conduction of water.

(D) Phloem transport is "bidirectional," meaning it can move food both upward and downward.

23. Bee-keeping (Apiculture) is often done alongside crop production as a "dual-purpose" venture. Why is this integration beneficial?

(A) Bees act as efficient pollinators, which significantly increases the yield of many fruit and oilseed crops.

(B) The bees utilize the nectar and pollen from the crops to produce honey and beeswax, providing extra income.

(C) The bees consume the harmful caterpillars and pests that attack the crop leaves.

(D) Bees produce "Royal Jelly," which can be used as a natural fertilizer to increase soil nitrogen.

Section - 3

COMPREHENSION-1

Forces govern interactions either through direct physical contact—like friction and tension—or across distances, such as gravitational and magnetic pulls.

Regardless of how the force is applied, a body's inherent inertia will resist any change to its state of rest or uniform motion unless a net unbalanced force acts upon it.

24. You are cleaning a rug by hanging it outside and beating it with a stick. what is the specific physical reason the dust separates from the carpet during this process?

(A) The beating creates an unbalanced electrostatic force that repels the dust away from the fabric.

(B) The beating sets the carpet in motion, whereas the dust particles tend to remain at rest due to the inertia of rest, causing them to separate.

(C) The force of friction between the dust and the carpet is perfectly balanced by the force of gravity, allowing the dust to fall.

(D) The carpet's inertia of motion transfers momentum directly to the dust particles, pushing them outward tangentially.

25. Three forces are acting simultaneously on a single, stationary point. if this system is in perfect equilibrium, what must be true about the relationship between these forces?

(A) Each force is proportional to the sine of the angle between the other two forces.

(B) The net external force acting on the system must be greater than zero to maintain the state of rest.

(C) The three forces must be entirely represented by the three sides of a right-angled rectangle.

(D) All three forces must be non-contact forces acting in the exact same direction.

26. Imagine you push a heavy box on the ground, but it does not move. You then push significantly harder, and the box finally starts sliding forward. what happened to the forces acting on the box at the exact moment it began to move?

(A) The force of gravity became greater than the normal reaction force of the ground, propelling the box forward.

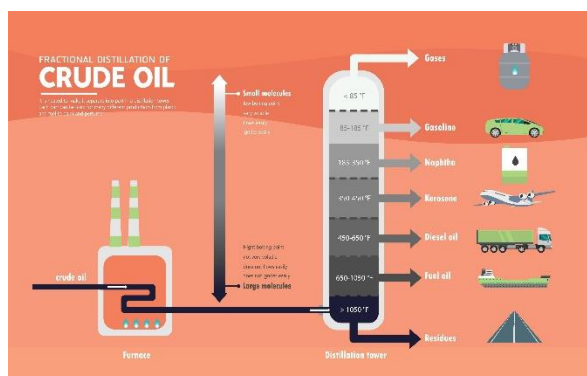
(B) The force of friction was completely eliminated, making the net resultant force zero.

(C) The forces transitioned from balanced to unbalanced because your push became greater than the opposing force of friction.

(D) The balanced forces simply changed the shape of the box instead of changing its state of rest.

COMPREHENSION-2

Crude oil (petroleum) is a "black gold" mixture of various hydrocarbons, ranging from very light gases to thick, heavy solids like asphalt. To make this mixture useful, it must be processed in an oil refinery using **Fractional Distillation**.



The process follows these steps:

Heating: The crude oil is heated to about 400°C (752°F) and pumped into the bottom of a tall fractionating column.

Vaporization: Most of the components turn into vapor and begin to rise through the tower.

Temperature Gradient: The tower is hottest at the bottom and gradually becomes cooler toward the top.

Selective Condensation: As the vapours rise, they cool down. When a specific hydrocarbon reaches a level where the temperature is just below its boiling point, it condenses back into a liquid and is collected on trays.

27. Based on the case study, which of the following substances would be collected at the lowest (hottest) part of the fractionating tower?

- (A) Petrol (Gasoline) for cars.
- (B) LPG (Liquefied Petroleum Gas) for cooking.
- (C) Lubricating oil and Bitumen for roads.
- (D) Kerosene for jet fuel.

28. Why is "Simple Distillation" insufficient for separating crude oil into its components?

(A) Simple distillation can only separate solids from liquids.

(B) The boiling points of the various hydrocarbons in crude oil are too close together.

(C) Simple distillation requires the mixture to be frozen first.

(D) Crude oil is a homogenous mixture, and simple distillation only works for heterogenous mixtures.

29. If a refinery technician finds that the gasoline (petrol) fraction is being contaminated with heavier kerosene, what is the most likely technical cause?

(A) The temperature at the top of the tower has become too low.

B) The temperature at the gasoline collection tray has become too high.

(C) The crude oil was not filtered for sand before heating.

(D) The pressure inside the tower has dropped to zero.

COMPREHENSION-3

A commercial beekeeper, Mr. Sharma, manages an apiary of *Apis mellifera* (Italian bee) situated next to a large apple orchard. In the spring, he notices that while his bees are active, the honey yield per hive has dropped by 30% compared to the previous year.

At the same time, the orchard owner reports that the apple trees are producing fewer fruits, even though they flowered abundantly. Upon investigation, a local agricultural scientist finds that the neighboring farm has shifted from growing mixed wildflowers and clover to a "monoculture" of a specific hybrid

sunflower that blooms only for two weeks in late summer. Additionally, the neighboring farm has increased the use of systemic insecticides to control beetles.

30. Based on the passage, what is the most likely primary cause for the simultaneous drop in honey yield and apple production?

- (A) The *Apis mellifera* species is naturally incompatible with apple blossoms.
- (B) A lack of "Pasturage" (floral variety and availability) throughout the spring season.
- (C) The bees have become "lazy" due to the high sugar content of the new hybrid sunflowers.
- (D) The apple trees require wind pollination, so the bees' presence is irrelevant to fruit set.

31. Why was *Apis mellifera* specifically chosen by Mr. Sharma for commercial honey production over the local *Apis cerana indica*?

- (A) It is a stingless variety, making it safer for the farmers to handle.
- (B) It has a high honey-collection capacity and stays in a given beehive for long periods.
- (C) It is the only species capable of surviving the systemic insecticides used on the neighboring farm.
- (D) It produces "Royal Jelly" that is chemically identical to the nectar of apple blossoms.

32. The most likely reason for the 30% drop in honey yield is:

- (A) Decrease in bee population
- (B) Lack of diverse flowering plants

throughout the season

- (C) Increase in apple tree production
- (D) Excess rainfall

Section - 4

33.

COLUMN A	COLUMN B
(A) Area under force–displacement graph	(P) Always non-zero over closed path
(B) Work done by conservative force in round trip	(Q) Represents total work done
(C) Work done by friction in complete cycle	(R) Independent of path
(D) Work done by force in different frames	(S) Depends on displacement measured

- (A) A–Q, b–P, c–R, d–S
- (B) A–P, b–Q, c–S, d–R
- (C) A–Q, b–R, c–P, d–S
- (D) A–R, b–P, c–Q, d–S

34.

COLUMN -A	COLUMN-B
(A) A smell of hot sizzling food reaches you several meters away.	(P) Particles have a fixed, closely packed arrangement with only vibratory motion.
(B) A diver is able to cut through water in a swimming pool.	(Q) High kinetic energy and large intermolecular spaces allow for rapid diffusion.
(C) A wooden table is called a 'solid' at room temperature.	(R) High pressure and low temperature reduce the distance between molecules.
(D) Liquefaction of atmospheric gases (like Nitrogen).	(S) Intermolecular forces are strong enough to hold volume but weak enough to allow fluidity.

- (A) A-R, B-P, C-Q, D-S
 (B) A-P, B-R, C-S, D-Q
 (C) A-R, B-P, C-S, D-Q
 (D) A-Q, B-R, C-P, D-S

35.

COLUMN -A	COLUMN-B
(A) Photosynthesis	(P) Ribosomes
(B) Protein Synthesis	(Q) Chloroplasts
(C) Steroid synthesis	(R) Lysosomes
(D) Waste disposal	(S) Smooth ER

- (A) A – P, B – Q, C – R, D – S
 (B) A – Q, B – P, C – R, D – S
 (C) A – R, B – S, C – Q, D – P
 (D) A – Q, B – P, C – S, D – R